# UDCT v7.08

Prior-art correction: the action-derived Hessian source term and screening by second derivatives of the Newtonian potential belong to Milgrom's generalized QUMOND (GQUMOND) class

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**Status.** Correction note; no new calculation. The prior-art surveys of v5.45–v5.46 cited Milgrom (2023) only for existence and uniqueness of solutions and missed that the same paper introduces the structures later used in O8c, O8d and v7.02–v7.06. This note records the omission and its consequences for how those results should be described. No verdict changes. This revision (before deposit) also corrects an overstatement about "negative" quantities in MOND (§3).

## 1. What GQUMOND contains

Milgrom (2023) generalizes QUMOND by letting the gravitational Lagrangian depend on higher derivatives of the Newtonian potential ψ: $L_G = -(1/8\pi G)\{2\nabla\varphi\cdot\nabla\psi - a_0^2 P[\nabla\psi, \nabla\nabla\psi, \ldots]\}$ [1]. The MOND potential then obeys a Poisson equation whose effective source includes a term $-\nabla_i\nabla_j(\partial P/\partial\psi_{,ij})$ (his Eqs. 7 and 14). This is the same double-divergence structure as the term $-\tfrac{1}{2}\partial_i\partial_j P_{ij}$ studied in UDCT v5.59–v7.06.

His first example makes P depend on $u = (\nabla\psi)^2/(\ell_0^2\psi_{,ij}\psi_{,ij})$ through $P = f(u)\, Q[(\nabla\psi)^2/(f\, a_0^2)]$, with $f(u) = u/(1+u)$ as an illustration, in order to screen MOND effects in systems smaller than a length $\ell_0$. He notes explicitly that this can greatly reduce the Galactic external-field anomaly in the inner Solar System when the solar balance scale (≈ 6000 AU) is below $\ell_0$, can suppress MOND effects in binaries and star clusters, and "possibly dwarf galaxies" [1]. He also remarks that the extra dimensioned constant can be replaced by a dimensionless multiple of MOND units built with c and $a_0$, necessarily very different from unity [1].

## 2. Consequences for the UDCT record

| Item | Correction |
|---|---|
| O8c/O8d (v5.57–v5.61) | These are GQUMOND-type theories: P depends on ∇ψ and on the Hessian through $X = c^2\tau/\lvert\nabla\psi\rvert^2$. The motivation (suppressing the solar external-field quadrupole by a Hessian-based screening) is anticipated by Milgrom (2023). The dimensionless X* built with c is an instance of his remark on dimensionless constants. Babichev et al. (2011) [4], cited in v5.57, is the more distant precedent. |
| v5.45–v5.46 prior-art verdicts | The entries "compactness screening: not found" and the novelty assessment of O8c must be read as superseded: the class exists. |
| v7.06 title and §6 | "A Hessian term absent from the standard MOND equation" is correct for QUMOND and AQUAL, but the term is present in the published GQUMOND class. Future versions must cite Milgrom (2023) as the direct precedent for the term, not only as a "comparison class". |
| Scope of the v7.06 negative work | UDCT built its action as $Q(z,\tau) = \int_0^z \nu_{\rm eff}\, dz'$ at fixed τ, so that the leading term reproduces a chosen $\nu_{\rm eff}$ exactly; its Hessian coefficient then integrates over dummy arguments reaching the screened regime (v7.01 §6). Milgrom's example instead uses an explicit P = fQ, whose Hessian term scales with the local logarithmic derivative of f and is small where f ≈ 1. Therefore the v7.06 obstruction must not be generalized to GQUMOND without computation; it may be specific to the integral construction. |

## 3. Related but distinct: negative phantom density in the MOND literature

Negative phantom densities are known in MOND: Milgrom (1986) asked whether the hidden mass can be negative, and in QUMOND the phantom density is negative in regions such as the neighbourhood of the midpoint between two masses [1]. These are local regions and reduce the Newtonian attraction there. The v7.05–v7.06 result concerns something related but larger: the angle-averaged radial force inside the stellar body of an ultra-faint dwarf is directed outward, with magnitudes 10–483 times the stellar Newtonian force (v7.06 Table 1). By Gauss' law an outward mean force means that the enclosed effective mass (stars plus phantom) is negative. The earlier statement that this is a wholly different kind of negative is withdrawn: the two share the same origin, an effective source of negative sign, and differ in extent and magnitude. Whether QUMOND or GQUMOND can make the enclosed effective mass negative in a strong external field was not established by the literature read for this note; it is registered as a diagnostic in v7.09.

## 4. Next step

Whether Milgrom's own example shares the v7.06 obstruction is an open, well-posed question. It is pre-registered separately (v7.09) before any calculation.


## Acknowledgement of assistance

Literature search and drafting of this note were carried out with the assistance of a large language model (Anthropic Claude) under the author's direction; the cited paper was read directly (arXiv:2305.01589).


## References


[1] M. Milgrom, Generalizations of quasilinear MOND, Phys. Rev. D 108, 084005 (2023), arXiv:2305.01589.

[2] M. Milgrom, Quasi-linear formulation of MOND, MNRAS 403, 886 (2010), doi:10.1111/j.1365-2966.2009.16184.x.

[3] M. Milgrom, General virial theorem for modified-gravity MOND, Phys. Rev. D 89, 024016 (2014).

[4] E. Babichev, C. Deffayet and G. Esposito-Farèse, Improving relativistic MOND with Galileon k-mouflage, Phys. Rev. D 84, 061502 (2011).

[5] P. J. McMillan, MNRAS 465, 76 (2017), doi:10.1093/mnras/stw2759.

[6] A. B. Pace, Local Volume Database (2025), doi:10.33232/001c.144859.

[7] UDCT v5.45–v5.46 (prior-art notes), v5.57–v5.61 (O8c/O8d; v5.58 doi:10.5281/zenodo.22987510, v5.60 doi:10.5281/zenodo.22994692), v5.64, v7.01–v7.06 (v7.06 research record and released script).